THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

Dec 14, 2018 TIME: 9:30 a.m.-12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL TEN questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.
- Some formulae you may need:

For $n \geq 2$:

$$\int \cos^n \theta \, d\theta = \frac{1}{n} \cos^{n-1} \theta \sin \theta + \frac{n-1}{n} \int \cos^{n-2} \theta \, d\theta$$

$$\int \sin^n \theta \, d\theta = -\frac{1}{n} \sin^{n-1} \theta \cos \theta + \frac{n-1}{n} \int \sin^{n-2} \theta \, d\theta$$

1. (10 marks) Give answers only to the following questions. The process of calculating is **NOT** required.

   (a) (2 marks) Find the value(s) of h for which the following set of vectors $\{v_1 = (1,0,0)^T, v_2 = (h,1,-h)^T, v_3 = (1,2h,3h+1)^T\}$ is linearly independent.

   (b) (2 marks) For which choice(s) of the constant k is the following matrix invertible?

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & k \\ 1 & 4 & k^2 \end{pmatrix}.$$

   (c) (6 marks) A is an $n \times n$ matrix. $|A| = x$, the rank of A is y and p is one of the eigenvalues of A. If we double every entry of A and get a new matrix B, find

   (i) the determinant of B.

   (ii) the rank of B.

   (iii) one eigenvalue of B.

2. (10 marks) For each of the following statements, write (T) if it is true and (F) if it is false, and then give a brief (say, one-line) explanation. Each question carries 2.5 marks.

   (a) A linear system of four equations in three unknowns is always inconsistent.

   (b) A linear system with fewer equations than unknowns must have infinitely many solutions.

   (c) If the system $Ax = b$ has a unique solution, then A must be a square matrix.

   (d) A homogeneous system is always consistent.

3. (10 marks) Given that

$$A = \begin{bmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

(a) (5 marks) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) (2 marks) Find a non-singular matrix P and a diagonal matrix D, such that $A = PDP^{-1}$.

(c) (3 marks) Find the $(3,3)$-entry of A^{2018}.

4. (10 marks) Determine whether the function

$$f(x,y) = \begin{cases} \dfrac{2x^2y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

5. (10 marks) Find the absolute minimum and absolute maximum value of the function $f(x,y) = \sqrt{x^2+y^2-2x+2}$ on the closed half disk $R = \{(x,y) : x^2+y^2 \leq 4, y \geq 0\}$.

6. (10 marks) Find the area inside the cardioid $r = 1 + \sin\theta$ and outside the circle $r = 3\sin\theta$.

7. (10 marks) Using ONE iteration of Newton-Raphson's method with initial guess $(0.5, 1.5)$ to find an approximation to the critical point of $f(x,y) = x^x \sin y$. (Round your solution up to 4 decimal places.)

8. (10 marks) Evaluate $\displaystyle\iint_R \frac{y}{x}\,dx\,dy$, where R is the region in the xy-plane bounded by the curves $x^2 - y^2 = 1$, $x^2 - y^2 = 4$, $y = 0$ and $y = \frac{x}{2}$. (Keep EXACT solution.)

9. (10 marks) Use the method of Lagrange multiplier to find the area of the intersecting surface of $x^2 + y^2 + 4z^2 = 1$ and $x + y + z = 0$. (Hint: the area of an ellipse equals to πab, where a and b are the semi-axes of the ellipse.)

10. (10 marks) Given that the density at any point is proportional to its distance from the center of the base, find the mass of a solid hemisphere H with radius a.

    (a) Write the integral in cylindrical coordinates.

    (b) Write the integral in spherical coordinates.

    (c) Evaluate any ONE of the integral in part (a) and (b).

******* **END OF PAPER** *******